Course Information

CS 111-02 - Foundations of Program Design
Instructor: Dr. Jasmine Tan Otto (she/her) jtotto@usfca.edu
TA: Priya Akhtarmoghaddam pakhtarmoghaddam@dons.usfca.edu
TA: Enzo Queirolo ejqueirolo@dons.usfca.edu

I'll respond to e-mail and Slack in 24 hours, except over the weekend.
If you message me on Canvas or on Discord, no guarantees.

Dates: Aug 25th – Dec 9th. Times: 1:10 pm - 2:55pm on Mondays and Wednesdays.

The final exam will be held 3pm on Fri, Dec 11th.
We will not meet on Labor Day (Sep 7th) or Fall Break (Oct 19th).

Text #1: Head First Java, 3rd Edition, by Kathy Sierra, Bert Bates, and Trisha Gee https://learning.oreilly.com/library/view/head-first-java/9781492091646/
Text #2: Learning Processing by Daniel Shiffman, as The Coding Train https://thecodingtrain.com/tracks/learning-processing

You can find the 1st text through the library, and to the 2nd through YouTube. Please let me know if you have any difficulty accessing these resources.

Office Hours

Jasmine jtotto@usfca.edu
Mondays 3:30pm - 4:30pm, HR 407B. Wednesdays 11am - 12noon, HR 407B.

Paria pakhtarmoghaddam@dons.usfca.edu
Thursdays 12:30 - 1:30pm, HR 411/413 (Tutoring center). Fridays 2 - 3pm, HR 411/413 (Tutoring center).

Enzo ejqueirolo@dons.usfca.edu
Wednesday 3:30 - 5:30pm. Saturday 12 - 2pm on Zoom, by appointment.

Resources

Find this document on GitHub with the latest information:
https://github.com/CS111-jtotto/Course-Information-Fall2026

The CS tutoring center is located on the 4th floor of Harney Science. You are welcome to drop in and get help with CS 111 assignments.

Graded Assignments

Attending section is mandatory. Talk with me ahead of time if you need to be excused. Topics that are discussed in class but are not available online will be part of quizzes, assignments, and examinations.

Labs (20%, participation)

You will write programs in-class based on exercises distributed during lecture, through the local router. While you're in the classroom, you will connect to the local router, which does not have internet access. The instructor will be available to help you debug problems with your code.

Lab assigments are given on Mondays; on Wednesdays, they're replaced by pair-programming. There are ten lab assignments.

Your lowest two lab grades will be dropped. If you need to miss a class, you'll get a zero for that lab.

Projects (20%, participation)

You will write programs outside of class based on instructions available on GitHub. You will have access to coding agents such as Claude and Gemini, and are welcome to use them for debugging. Coding agent use **must be disclosed**, however.

LLM use is not mandatory to complete any assignments. You will have an easier time understanding your code if you write it in small parts, which is easiest by hand.

There are three projects, and you will have three weeks to complete each one.

Keep in mind that quizzes and exams will require you to write code by hand, and reason about its behavior. If you write a lot of code, but cannot read it to your own satisfaction, then you will miss out on an opportunity to practice the skills we are learning.

Quizzes (30%, graded)

Quizzes based on the previous week's material will be given in-class on Wednesdays at the start of class. You will have 20 minutes to complete 2 - 3 programming questions on a sheet of paper.

Quizzes are open notes. You may use any amount of written or printed notes or reference material. You may not use your phone or your laptop. If you want to reference the electronic course text, I recommend printing or copying out its material ahead of time.

Quizzes are given on Wednesdays; on Mondays, they're replaced by a debugging match. There are ten quizzes.

Your lowest two quiz grades will be dropped. If you need to miss a class, you'll get a zero for that quiz.

Midterm Exam (15%, graded)

The midterm exam will cover similar material to the quizzes in the first half of the course. It will be administered on paper, taking the full duration of section. You may bring any amount of written or printed notes or reference material.

You are welcome to prepare for the exams by reviewing your graded quizzes, working on the labs and projects, and by asking questions in review sessions we will hold during lecture.

Final Exam (15%, graded)

The final exam will cover similar material to the quizzes in the second half of the course. It is similar to the midterm exam in its form and method of evaluation.

If you cannot make the final exam at 3pm on Friday, Dec 11th, let me know by Monday.

Grade Assignment

Your final grade is based on your graded materials (quizzes, exams) and completed assignments (labs, projects).
Each category is weighted as indicated above.

grade	total score
A	>= 93%
A-	>= 90%
B+	>= 87%
B	>= 83%
B-	>= 80%
C+	>= 77%
C	>= 73%
C-	>= 70%
D	>= 60%
F	< 60%

Submission Policy

Upload labs by the end of class through the router. No late submissions will be taken; you have to do lab work in-class.

Upload all project materials by 11:59pm on the due date to GitHub. If a submission is late, 20% of the score will be deducted for each day it is late. No e-mail submissions.

Quizzes and exams cannot be retaken. If you will miss an exam day, talk with the instructor in advance.

Academic Dishonesty

Students are required to follow the University's Honor Code:

> As a Jesuit institution committed to cura personalis - the care and education of the whole person - USF has an obligation to embody and foster the values of honesty and integrity. USF upholds the standards of honesty and integrity from all members of the academic community. All students are expected to know and adhere to the University's Honor Code.
> You can find the full text of the code online: www.usfca.edu/fogcutter

tl;dr: Don't represent the work of other people as your own work.

If you have any doubt about how you may discuss an assignment with other people, ask me about it.

- You're welcome to get help from lab tutors, the course TAs, and the instructor on coding assignments. If you wish to get a tutor for this course, talk with me. Take advantage of the school's resources.
- Let me know before you ask for help from a classmate, an outside tutor, or a family member. You must disclose assistance that you receive from any source other than the instructor and the TAs.
- Disclose your use of coding agents. I want you to get better at debugging, not copy-pasting.

Violations of academic honesty will result in severe penalty. A first offense will result in a zero on the assignment and a report to the Dean's office. A second offense will result in you failing the course.

Schedule

Week 1. Wed: Intros. Review syllabus, discuss course expectations. Install OpenJDK, Git, and Sublime Text.

Week 2. Mon: Classes, `main()` method, and `System.out.println()`. Lab assignment: Hello World.
Wed: First quiz. Using `int` and the = assignment operator. Variables vs literals.

Week 3. Mon: No class (Labor Day). Wed: The == equality operator, if/else statements, and variable scope. Writing comments.

Week 4. Mon: While loops and for loops. Return types. Lab assignment: FizzBuzz.
Wed: Boolean operators, floating point arithmetic. How to override type checking.

Week 5. Mon: Helper methods, try/catch blocks, `java.io.BufferedReader` usage. Lab assignment: CSV parser.
Wed: Fixed-length arrays. String and StringBuilder. Assign first project.

Week 6. Mon: Midterm review, no lab due.
Wed: Midterm exam.

Week 7
Mon: Fields, constructor methods, and the `new` keyword. Install the Processing IDE. Lab assignment: Project 1 check-in.
Wed: Return graded midterms. No quiz. Instances vs primitives. Common ways of encountering a `NullPointerException`.
Last week to drop.

Week 8
Mon: Guest lecture, no lab due.
Wed: Guest lecture, no quiz. Submit first project.

Week 9
Mon: No class (Fall Break)
Wed: Review constructors and fields. `init()` and `loop()` methods. Assign second project.

Week 10
Mon: `pushStyle` and `popStyle`. Mouse input, event handlers. Lab assignment: Project 2 check-in.
Wed: For-each loops, `java.util.ArrayList`, primitive wrappers. Strategies for reading API documentation.

Week 11
Mon: Coordinate transformations, `pushMatrix` and `popMatrix`. Lab assignment: TBA
Wed: Fractals and recursion. `java.util.Random` and Monte Carlo simulation.
Last week to withdraw.

Week 12
Mon: `java.util.Stack` and `java.util.Set`. Install IntelliJ IDEA. Lab assignment: TBA
Wed: Library imports. Static variables and methods. No quiz. Submit second project, and assign third project.

Week 13
Mon: Key-value pairs, `java.util.HashMap`. Project file structure. Lab assignment: Project 3 check-in.
Wed: `extends` and the `super` constructor. Overriding methods.

Week 14
Mon: `PrintWriter` and file outputs. Review file inputs. Lab assignment: TBA
Wed: Exception handling, object polymorphism.

Week 15
Mon: Review control flow and return types. Maps and filters. Lab assignment: TBA
Wed: `ArrayList.sort` and the `Comparator` interface. Accumulators and reducers. Submit third project.

Week 16
Mon: 2D arrays and nested loops. Strategies for evaluating time complexity.
Wed: Final review.

Final Exam: Friday, Dec 11th, 3 - 5pm.